

EDUCATION

Ohio State University (OSU)

B.S. in Mathematics, Minor in Philosophy

Summa Cum Laude, with Honors in the Arts and Sciences

Aug 2022–Dec 2024

Cum GPA 3.934/4.0

RESEARCH EXPERIENCE

Analyzing self-coherence of stochastic oscillations emitted by ears

Jan 2025–Present

Supervisors: Prof. Christopher Bergevin; Prof. Natasha Mhatre

Sensory Biophysics Lab, York University

Developed signal processing methods to quantify properties of faint sounds recorded from ears

- Lead author of [bioRxiv preprint](#); will be presenting at Association for Research in Otolaryngology 2026 MidWinter Meeting
- Released Python package [phaseco](#) implementing our methods; integrated analyses into a [unified codebase](#)
- Engineered dynamic windowing techniques to address an issue related to time–frequency uncertainty

Fields Undergraduate Summer Research Program

Jun 2024–Aug 2024

Supervisors: Prof. Christopher Bergevin; Prof. Natasha Mhatre

The Fields Institute

Studied oscillatory emissions from lizard ears during funded research program

- Simulated locally coupled oscillator models of these emissions and investigated role of amplitude-dependent frequency in synchronization
- Examined how Volterra-series methods can be extended to handle stochastic forcing of nonlinear oscillators
- Expanded existing model to incorporate coupling between lizard ears

Topological data analysis of serotonin time-series data

Jan 2024–May 2024

Supervisor: Prof. Janet Best

OSU

Topological data analysis approach to find differences in serotonin time-series data from depressed vs control mice

- Designed algorithm for identifying release/clearance processes building on sublevel set filtration-based method
- Discovered trend suggesting the measurement itself may have disproportionately affected depressed mice

Optimizing an efficient eigenvector phase retrieval algorithm

Jan 2023–May 2023

Cycle Undergraduate Research with Daniel Packer (PhD Candidate)

OSU

Designed and implemented method for efficiently recovering eigenvectors from the absolute value of their entries

PREPRINTS

Peacock, S. N. S., Vencovsky, V., Whiley, R. E., Mhatre, N., & Bergevin, C. (2025). *Spontaneous oto coherence of the active ear.* bioRxiv. <https://doi.org/10.1101/2025.11.14.687084>

SELECTED PROJECTS

[phaseco](#)

A Python package providing tools for phase autocoherece analysis with dynamic windowing methods

[Peak-picking otoacoustic emissions with machine learning](#)

A neural network model to identify features in power spectra of oscillations emitted from the ear

- Designed network by integrating approaches from other domains; generated synthetic data for supervised learning

[Quantifying uncertainty in college basketball](#)

Devised Elo-style ranking to derive a bootstrapped C.I. for the minimum “uncertainty” in a game

- Top OSU team at TriCOMM Mathematical Competition in Modeling, November 15th–17th 2024

PRESENTATIONS

Association for Research in Otolaryngology MidWinter Meeting	Feb 2026
<i>Otocoherence: Interspecies analysis of phase self-consistency in spontaneous otoacoustic emission</i>	
OSU Honors Project Symposium	Dec 2024
<i>Topological data analysis of serotonin concentrations from depressed mice</i>	
Fields Undergraduate Summer Research Program Final Presentations	Aug 2024
<i>Nonlinear dynamics of otoacoustic emissions</i>	

HONORS AND AWARDS

Conference Travel Award for the Association for Research in Otolaryngology 2026 MidWinter Meeting	2025
Top OSU Team at TriCOMM Mathematical Competition in Modeling	2024
OSU Maximus Scholarship (All Semesters)	2022–2024
OSU Dean’s List (All Semesters)	2022–2024
National Merit Scholar Finalist	2018

SKILLS AND COURSEWORK

OSU Coursework

Mathematics: Real analysis I & II, linear algebra, ODEs, PDEs, probability, dynamical systems

Applied Mathematics: Quantitative neuroscience, statistics, machine learning, infectious disease dynamics

Physics: Classical mechanics, electricity and magnetism, relativistic mechanics, quantum mechanics

Audited Coursework & Independent Study

Complex analysis, functional analysis, applied algebraic topology, discrete-time signal processing

Computational Skills

Signal processing, machine learning, topological data analysis, package development

Languages: Python (advanced); MATLAB, Java (intermediate)

TEACHING / MENTORING

Private mathematics tutoring	Aug 2025–Present
Tutored OSU undergraduate student in real analysis and proof techniques	

Mentoring during Fields Undergraduate Summer Research Program	Jun 2025–Aug 2025
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Mentored Prof. Bergevin’s group of undergraduates in later iteration of research program

- Proposed accessible entry points based on students’ interests into our research modeling, simulating, and analyzing stochastic oscillations emitted by ears
- Led regular student meetings for troubleshooting and sharing progress

Private music lessons	Jun 2018–Present
Taught lessons in music theory, guitar, and bass	

OTHER EXPERIENCE

Cyclops Studio and Effects: Effect Pedal Technician, Studio Engineer	Jul 2019–Present
Repaired, designed, and built electronic musical effect units; engineered and mixed recordings	

Rat Motel: Band Manager, Songwriter, Performer	Jun 2016–Present
Booked tours, managed merchandise, and kept financial records, developing project management and organizational skills	